

GUIDELINES ON THE PROVISION OF BUFFER ZONE AROUND WASTE PROCESSING AND DISPOSAL FACILITIES



Central Pollution Control Board
April, 2017

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1. Introduction

Indian cities are expanding with the increase in population, economic activities and the resulting urbanization. Whereas population residing in urban areas was 11.4% of total population in 1901, it increased to 28.53% in the 2001 census and crossed 30% as per 2011 census, standing at 31.16%. There are 53 urban agglomerations in India with a population of 1 million or more as of 2011 against 35 in 2001. About 43 percent of the urban population of India lives in these cities. The unprecedented growth of these cities has posed several challenges for municipal authorities. Identification of suitable sites for waste management infrastructure in cities is one of the toughest challenges municipal authorities are facing at present. Lack of proper/ updated land use plan with urban authorities is a stumbling block in implementing solid waste management projects.

Most of the existing solid waste management facilities are practicing crude dumping of solid waste. In some cases where solid waste is processed, the situation is still alarming due to use of conventional treatment technologies coupled with poor operation and maintenance by the fund starved ULB. This situation is giving rise to numerous environmental and public health concerns in and around urban areas. “Not in My Back Yard (NIMBY) syndrome” and litigations are common as public at large do not trust ULBs in providing credible waste management services. Majority of existing solid waste treatment plants and dumping sites, though initially away from habitation but now have no adequate buffer zone from these habitations. Buffer even where available have come under illegal encroachment in many cities and settling societies demand shifting the waste treatment facility itself. Thus there is a general public resistance to the location of waste management facility in any area. Lack of identified sites for municipal solid waste management in master plan compounds the problem.

Disposal of waste in landfills/ dumpsites without any treatment is still practiced even as it impacts on the surrounding environment. Waste management sites encompass waste processing/disposal facilities, which become sources of pollution in terms of air, water, land and noise besides emitting foul smell. Therefore, provision of buffer zone around these facilities is essentially required to protect people living in the surroundings from

exposure/impacts of such pollutants but also to ensure continued safe operations in the waste management facility by maintaining its “island character”. Buffer zone also acts as barrier, absorber and to some extent as remedial measure against the fugitive emissions. Fugitive emissions of pollutants emitted during handling of waste, storage, transportation and movements of traffics.

Currently, no scientific basis is available for making provisions for buffer zone around waste processing/disposal facilities. The provisions recommended in the “Municipal Solid Waste Management Manual, 2016” were broadly drawn from the “Report of the Committee constituted by the Hon. Supreme Court of India in March 1999” on Solid Waste Management in Class 1 Cities in India.

In this context, the Government of India through CPCB has framed these guidelines on maintaining Buffer zone including green belt around waste management facilities. These guidelines will not only facilitate the ULBs in meeting the regulatory requirements, reduce the aforesaid nuisance value of the waste management facilities but also make an effort to enhance their aesthetic appeal. In addition to above, the siting criteria for setting up these facilities for waste processing/ landfill is adopted as mentioned in SWM Rules, 2016 at tailing part of these guidelines.

In some instances, the actual separation distance may vary from those recommended in these Guideline, due to site-specific constraints. In such cases, variations to the recommended separation distances may be acceptable, subject to detailed assessment by concerned authorities and to the satisfaction of the State Pollution Control Board/Committee.

2. Objective of the Guidelines

The purpose of this Guideline is to specify adequate separation distances between solid waste management facility and its surrounding area having different land usage characteristics.

To achieve the purpose, these Guidelines aim to:

- minimize the risk of adverse impacts on the environment (land, air, water, noise pollution) and the impacts on the Public Health
- inform and support strategic land use planning decisions and prevent encroachment of controlled areas
- Generate/ develop public acceptance for solid waste treatment and disposal infrastructure
- Encourage new technological innovations for processing facilities with minimal land requirement

3. Regulatory Framework

The buffer zone was first envisaged in 1982 after Indian task force developed the 'Core-Buffer-Multiple Use Zone' strategy. This strategy aimed at separating incompatible land uses, particularly in relation to wildlife. In this approach, the buffer zone would be under the wildlife park authorities' administration and controlled use of forest produce would be allowed. The multiple-use zone was located outside the park boundaries designated for rural development. With similar analogy, these buffer zone guidelines are framed for waste processing and disposal facilities. The existing regulatory provisions for these guidelines are given as under:

- i. Provisions related to Buffer Zone specified in the **Solid Waste Management Rules, 2016** mentioned as under;
 - **Rule 11** Section (l)- **Duties of the Secretary-in-charge, Urban Development in the States and Union territories-** Notify buffer zone for the solid waste processing and disposal facilities of more than five tonnes per day in consultation with the State Pollution Control Board
 - **Rule 12** Section (h)- **Duties of Central Pollution Control Board-** Publish guidelines for maintaining buffer zone restricting any residential, commercial or any other construction activity from the outer boundary of the waste processing and disposal facilities for different sizes of facilities handling more than five tonnes per day of solid waste;

- The **distance/siting criteria's for setting up waste management facilities** as specified in Solid Waste Management Rules, 2016 at **Schedule I (A)(vii)**
 - **Schedule I (A) (viii)**-The sites for landfill and processing and disposal of solid waste shall be incorporated in the Town Planning Department's land-use plans.
 - **Schedule I (A) (ix)**-A buffer zone of no development shall be maintained around solid waste processing and disposal facility, exceeding five tonnes per day of installed capacity. This will be maintained within the total area of the solid waste processing and disposal facility. **The buffer zone shall be prescribed on case to case basis by the local body in consultation with concerned State Pollution Control Board.**
 - **Schedule I (F)**-Criteria for ambient air quality monitoring
- ii. The **Coastal Zone Regulation** notified by Ministry of Environment Forest And Climate Change also prohibits setting up and expansion of units or mechanism for disposal of wastes in High Tide Line (hereinafter referred to as the HTL) to 500 mts on the landward side along the sea front. Also dumping of city or town wastes including construction debris, industrial solid wastes, fly ash for the purpose of land filling and the like with high tide line shall be regulated by the concerned authority, where shall implement schemes for phasing out any existing practice, if any.
 - iii. The buffer zone guidelines for setting up processing and disposal facility also come under the purview of The Water (Prevention and Control of Pollution) Act, 1974, The Air (Prevention and Control of Pollution) Act, 1981.
 - iv. For setting up solid waste processing and disposal facilities, The Environment (Protection) Act, 1986 also need to be adhered to particularly from the angle of Environmental Clearances. Authorities concerned need to deliberate on the number of issues and criteria when siting a buffer zone as broadly categorized below:

a) Environmental considerations

- Distance from the flood plains, coastal regulation, wetland, Critical habitat areas, sensitive eco-fragile areas, highways, habitations, public parks and water sources

- Topography- Hilly areas, land availability and also the slope's landslide potential.
- Wind Speed and Direction- Wind direction is one of the important consideration as to the area that can be affected due to dust and odour.

b) Proximity and access considerations

- Transportation Network
- Utilities and Services

c) Land-use considerations

- Land Usage and Activities on Adjacent Sites
- Allowable Land Uses and Zoning
- Proximity to Airports
- Proximity to Other Waste Management Facilities

4. Existing Norms for Buffer Zone in India and Abroad

A.) Buffer Zone

The buffer zone, particularly in context of NIMBY syndrome in India, is one of the limiting conditions for obtaining Environmental Clearance for setting up solid waste processing and disposal facilities. At present, there are no published norms for buffer zone for solid waste management facilities by MoEFCC/ CPCB.

However, the “Manual on Municipal Solid Waste Management, 2016” published by CPHEEO, Ministry of Urban Development recommends certain provisions for buffer zone particularly the one of maintaining 500 m buffer zone around the waste processing facilities. In the given pace of urbanization in the country, getting such large piece of land is becoming increasingly difficult and costly. ULBs in setting up waste processing and disposal facilities expeditiously.

The provisions made for Buffer zone for solid waste processing and disposal facilities in various countries are tabulated below:

i. Landfill

International Solid Waste Association	500 m should be provided depending on the size of landfill, height, wind direction
South Australia	500m buffer distance shall be maintained between areas dedicated for waste disposal and the nearest surface water
Ontario, Canada	Buffer area shall be at least 100 m wide at every point, if that does not apply to a buffer area, if the buffer area is at least 30 metres wide at every point and a written report confirms that; (a) the buffer area provides adequate space for vehicle entry, exit, turning, access to all areas of the site and parking; (b) the buffer area provides adequate space on the surface of the site for all anticipated structures, equipment and activities; and (c) the buffer area is sufficient to ensure that potential effects of the landfilling operation do not have any unacceptable impact outside the site.
Malaysia	500m
South Africa	Buffer zone min 200m to 500m
Bangladesh	250m from the habitat
Hong Kong	250 m away from the edge of the waste (landfill boundary)

ii. Waste processing facilities

Canada	minimum buffer strip between composting facility boundary and adjacent property. For in-vessel Composting distance between active area and the nearest residential or institutional building shall be min 500m, nearest commercial or industrial building 250 m and nearest property boundary will be min 100m.
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CANADA-Nova Scotia	In case of in-vessel composting facilities, where it can be demonstrated that particular equipment will not release odours generated from the composting process into the surrounding environment, the distance between the equipment and the nearest property boundary shall be a minimum of 30 metres
Malaysia	production of compost from organic waste- 500m
Devon city Council (UK)	buffer distance 500m
China	300m buffer zone between incineration plants and local residents

From above, it is observed that the minimum buffer area varies from 100 m to 500 m in case of both waste processing and disposal facilities.

B.) Facility Siting Criteria

In addition to the suitable provisions of the buffer zone, the SWM Rules, 2016 provides norms for siting criteria for landfills. The same is reproduced below for adoption while setting up **landfill facilities**.

Table 1. Criteria specified for identifying Suitable Land for Sanitary Landfill Sites (Not a treatment facility)

S. No.	Place	Minimum Siting Distance
1.	Rivers	100 m away
2.	Ponds, Lakes, water bodies	200 m
3.	Highway, Habitations, Public Parks and water supply wells	200 m from center line
4.	Flood Plains as recorded for the last 100 years , zone of coastal regulation, wetland, Critical habitat areas, and sensitive eco-fragile	Sanitary landfill site not permitted

	areas	
5.	Airport/ Airbase	20 km**

In a special case, **landfill site** may be set up within a **distance of 10 and 20 km** away from the Airport/Airbase after obtaining no objection certificate from the civil aviation authority/ **Air force as the case may be.

However, there is no such siting criteria applicable for setting up waste processing facilities.

5. Recommended Provisions for Buffer Zone

The Solid Waste Management Rules, 2016 specified the terminology of **Buffer Zone**, as ***“no development zone to be maintained around solid waste processing and disposal facility, exceeding 5 TPD of installed capacity. This will be maintained within total land area allotted for the solid waste processing and disposal facility.”***

Buffer Zone around the core waste processing area consists of utility area, open parks and green belts etc. Further, depending on feasibility of planning, the interface land use between the boundary of waste processing facility and sensitive receptors, can also be developed as an additional measure. The layout of buffer zone (utility area, open parks and green belts) including core waste processing area and optional interface land use is shown in the figure below:

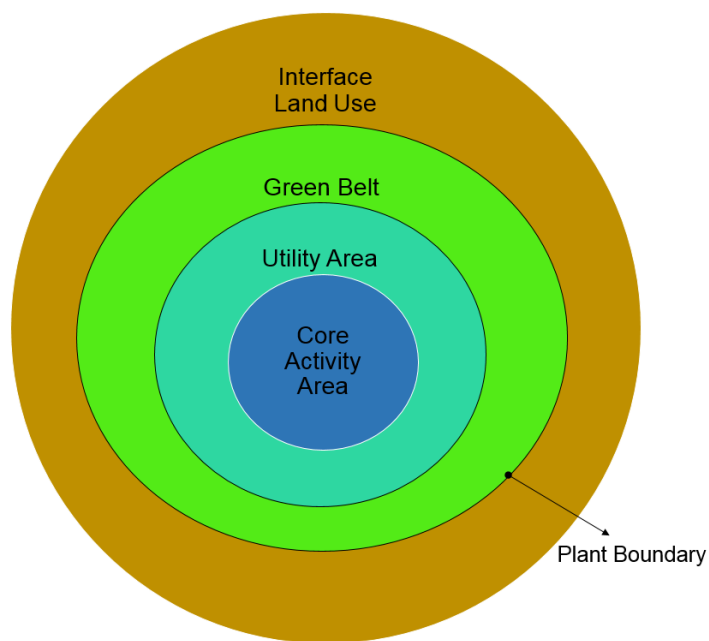


Figure 1 Depicts activity boundary, green belt and separation distance

For the purpose of these guidelines, the Buffer Zone, Separation Distance, Utility Area, Green belt and Interface Land use shall have the meanings set out below, unless otherwise provided, hereafter, for the exclusive interpretation of these Guidelines.

- a) The **Buffer Zone** is generally defined as an area of restricted activities, depending on the activity in adjacent land uses. It also ensures long-term continuous availability of disposal sites by avoiding potential conflicts between waste disposal sites and adjacent lands with different users.
- b) **Buffer Distance or Separation distance** is measured as the areal distance between the source of emission and sensitive receptors. For the purpose of these guidelines and addressing the required protection from adverse impacts, separation distance is measured from the tip of core SWM facility processing boundary, as the source of emission, to the nearest boundary of the property of sensitive receptors as shown in figure 1.

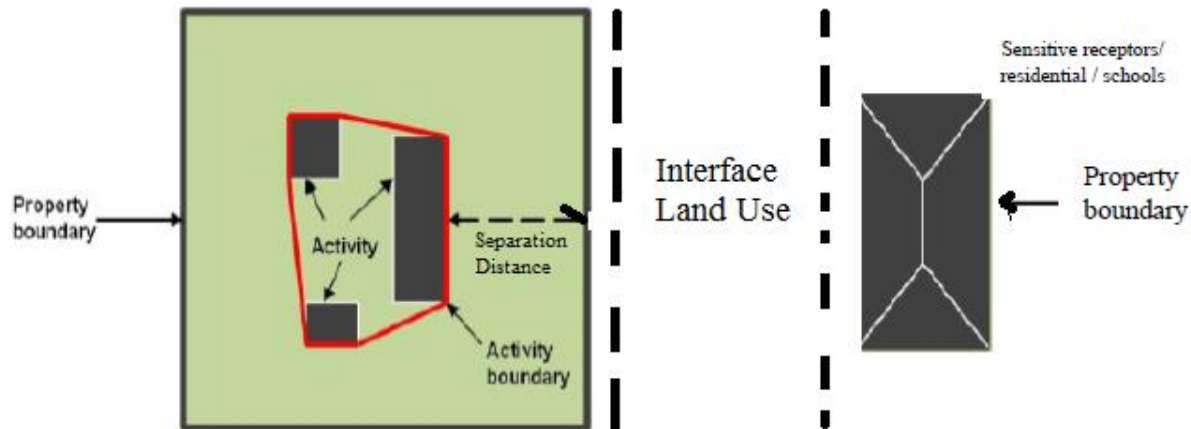


Figure 2. Core Plant activity area, buffer Zone and interface land use

- c) **Core Waste Processing/Landfilling Area** typically requires space for receiving waste, storing waste, segregation of waste and treatment units within the facility. Similarly, for Landfilling it is the area of cell which is receiving the waste/inert.
- d) **Utility Area** within the facility is designated area for the facility operations other than the core activities like. Weigh bridge, parking, vehicle cleaning, laboratory, emergency services etc.
- e) **Green Belt** for the purpose of these guidelines shall refer to an area that is kept in reserve within the allotted land for setting up facility, around the core SWM processing area, for the purpose of plantation and landscaping to reduce the adverse effects from pollutants like air & noise, soil erosion control etc. It also works as a natural shield to protect people around the facility from these pollutants.
- f) **Interface Land Use:** The buffer zone could be further augmented with interface land use area, where above beneficial and feasible as an additional optional measure, after due approval of the concerned authorities. The interface land use shall not generate significant emissions, nor warrants protection from them. The activities in the interface land use are **vehicle**

showrooms, service stations, warehouses, display homes, emergency services facilities, funeral, veterinary clinic and parks etc.

i. Separation Distances for Solid Waste Processing and Disposal Facilities

Ideally, a distance of 500 meter from the boundary of the Solid Waste Processing and Disposal Facility (sanitary landfill) should be maintained. However, on case to case basis a distance of minimum 200 meter from the Solid Waste Processing and Disposal Facility (sanitary landfill) can be considered subject to the condition that such facility meets the stipulated standards prescribed by State Pollution Control Board with respect to ambient air as well as for stack emissions.

The above provisions have been made keeping in view of high population density in urban areas, scarcity of land to set up such facilities and protest from local inhabitants in the area of processing/ disposal facility and is in line with those being adopted at international level. Besides, the following three conditions need to be ensured:

- (a) the buffer area provides adequate space for vehicle entry, exit, turning, access to all areas of the site and parking;
- (b) the buffer area provides adequate space on the surface of the site for all anticipated structures, equipment and activities; and
- (c) the buffer area coupled with technological interventions is sufficient to ensure that potential effects of the processing/ landfilling operation do not have any unacceptable impact outside the site.

6. Green Belt

The buffer zone effectiveness is reinforced by the green belt within the solid waste processing and disposal boundaries. An important aspect of a green belt sometimes overlooked is that the plants constituting green belts are living organisms with limits to their tolerance towards air pollutants. For the purpose of these guidelines, the green belt shall refer to an area that is kept in reserve within and around the SWM facility for the plantation and landscaping to reduce the adverse effects from the activity area like air &

noise pollution, soil erosion etc. The green belt is an effective pollution sink only within the tolerance limits of constituent plants. The philosophy is that when primary pollutants are taken care of, formation of secondary pollutants will not reach menacing proportions. Primary pollutants of concern are – SO₂, HF, NO₂, CO, CO₂, NH₃, H₂S, Cl, SPM and organics. **Annexure- 1** attached to these guidelines shows the selection criteria for plants near the processing facility.

These guidelines recommend minimum 10 metres green belt within and all around the facility along the boundary. Vegetation, shrubs, trees, and berms with high density greenery can be incorporated into green belt within facility limits to serve as visual barriers and to reduce noise levels. Depending on the monitoring of level of pollutants in ambient air after the boundary of facility, on case to case basis, suitable technological measures/ barriers to check pollutants need to be resorted. The important factors for developing green belt for agro-climatic conditions are stated below:

a) Criteria for Selection for Plant Species

- The plant species should be fast growing
- They should have thick canopy cover
- They should be perennial and evergreen
- They should have high carbon – CO₂ sink potential
- They should be effective in absorbing pollutants without significantly affecting their growth

b) Recommended plant species:

Keeping in view the nature of pollutants expected from the disposal site, a green belt of minimum 10 metre width is recommended and the following plant species can be selected for plantation:

- *Acacia nilotica* (Babul)
- *Deldergia Sissoo* (Shishum)
- *Acacia auriculiformis* (Australian Babul)

- Azadirachta Indica (Neem)
- Lagerstroemia speciosa (jamun)
- Prongamia pinnata (Karanji)

c) Recommended plant species Density around Processing & Disposal/ Landfill site:

These guidelines recommend the green belt width of minimum 10 meters within and all around processing and disposal facilities. The recommended minimum density of the green belt should be as discussed in the green belt model provided in the CPCB guidelines for developing green belts in 2000. These guidelines introduce the concept of a pollution attenuation coefficient for estimating the removal of pollutant while passing through the green belt. The formulation of pollution attenuation coefficient makes use of parameters such as leaf area, density of the tree plantation, deposition velocity of the pollutant on leaf surface and wind speed to the green belt. The model gives the dependence of the pollution attenuation factor of a green belt on various physical parameters of the green belt such as its height, width, distance from the pollution source and on atmospheric stability conditions and hence the model can be used to optimize the design of the green belt in obtaining the desired degree of attenuation of the pollution around an industry. The case to case basis CPCB guidelines for developing green belts (March, 2000) to be referred for optimal density applications.

7. Operationalization Framework

Solid Waste Management Rules, 2016 has empowered Central Pollution Control Board for maintaining buffer zones restricting any residential, commercial or any other construction activity from the outer boundary of the waste processing and disposal facilities for different sizes of facilities handling more than five tonnes per day of solid waste. The guidelines will be updated, from time to time, and address environmental aspects of processing and disposal of solid waste to enable local bodies to comply with the provisions of SWM Rules, 2016.

i. Role of State Pollution Control Board

- a) The SPCB shall link the buffer zone achievement with grant of Consent to operate and establish under stipulated norms;
- b) The SPCB shall conduct periodic environmental monitoring around buffer zone and assess the impact on the sensitive receptors;
- c) The SPCB shall bi-annually review the Green Belt condition within the facility premises and give suggestions to the ULBs for further improvements. Stringent measures and penalties as per the stipulated norms to be imposed in case of default;
- d) The SPCB shall extend all necessary support to local authority for the site selection for the newly proposed waste processing and disposal facility;

ii. Role of Local Body/ Facility Operator

- a) The ULB shall be responsible for the selection of site in close coordination with SPCB;
- b) The ULB/ operator shall be responsible for green belt development and maintenance in the buffer zone;
- c) The ULB shall direct the operator concerned, in case it outsources facility to comply with these guidelines

iii. Role of Town and Country Planning Department

- a) Town and Country Planning Department shall allocate adequate land for waste management facilities in the Master Land Use Plan;
- b) Town and Country Planning Department shall make all efforts to restrict/ prohibit peri-urban growth near such facility;
- c) Town and Country Planning Department shall be responsible for making provisions of Green Area development around such existing/ exhausted facilities to the extent feasible to minimize the impact of pollution to sensitive receptors.

8. Annexure-1- Selection criteria for plants near the processing facility

Table 2.6 Compilation of research in India indicating sensitive and tolerant species, with reference to industrial pollutants

Name of Plant	Sensitive	Tolerant	Reference
<u>Mangifera indica</u>	Coal dust		
<u>Citrus lemon</u>		Coal dust	Rao, 1971
<u>Phasaeolus aureus</u> (Green gram)	Petro cake		Prasad and Rao (1981)
<u>Zea mays</u>	Cement dust		Sree Rangaswamy <i>et al.</i> , (1973)
<u>Syzygium cumini</u>	Cement dust		Jafri <i>et al.</i> , (1979)
<u>Psidium guajava</u>	Cement dust		Yunus and Ahmed (1980)
<u>Triticum aestivum</u>	Cement dust		Singh and Rao (1980 a)
<u>Calotropis procera</u>	Cement dust		Yusuf and Vyas (1982)
<u>Cassia fistula</u>	Cement dust		
<u>Dalbergia sissoo</u>	Cement dust		
<u>Withania somnifera</u>	Cement dust		
<u>Glycine max</u>	Cement dust		Singh and Rao (1978 a)
<u>Hordeum vulgare</u>		5% fly ash	Bhatia (1978)
<u>Portulaca sp.</u>			
<u>Triticum aestivum</u>	above 20% Fly ash		Pawar and Dubey (1982)
<u>Triticum aestivum</u>		6g/m ² /day fly ash	Dubey <i>et al.</i> , (1982)
		4g/m ² /day fly ash	
<u>Dolichos lablab</u>		4g/m ² /day fly - ash	Pawar <i>et al.</i> , (bean) (1983)
<u>Abelmoschus esculentus</u>	Cement and	fly - ash	Pawar <i>et al.</i> , (1982)
Var Pusa savani	Coal dust		
<u>Commelina benghalensis</u>	Air borne dust		Chaphekar <i>et al.</i> , (1980)
<u>Brassica oleraceae</u>	Urban air		Garg and Varshney (1980)
<u>Chenopodium album</u>			
<u>Cicer arietinum</u>			
<u>Dolichos lablab</u>			
<u>Sonchus asper</u>			
<u>Withania somnifera</u>			
<u>Tabernaemontana coronaria</u>	Polluted environment		Srivastava <i>et al.</i> (1980)
<u>Calotropis procera</u>		Polluted conditions	Yunus and Ahmed (1981)

(Contd.....)

Table 2.6 (Contd.....)

Name of Plant	Sensitive	Tolerant	Reference
<u>Calotropis gigantea</u>	Polluted areas		Bhirava Murthy and Kumar (1983)
Baro paddy. Var. Ratna	Urban dust		Das and Pattanayak (1978)
<u>Mangifera indica</u>		Dust Collector	Shetye and Chaphekar (1980)
<u>Thespesia populnea</u>			-----
<u>Erythrina indica</u>	Poor dust Collector		
<u>Polyalthia longifolia</u>			
<u>Ficus benghalensis</u>		Dust Collector	Das 1981 and Das <i>et al.</i> (1981)
<u>Ficus infectoria</u>			
<u>Ficus religiosa</u>			
<u>Mangifera indica</u>			
<u>Tectona grandis</u>			
<u>Polyalthia longifolia</u>			
<u>Shorea robusta</u>			
<u>Terminalia arjuna</u>			
<u>Cassia fistula</u>	Poor dust Collector.		Das (1981) and Das <i>et al.</i> (1981)
<u>Poinciana regia</u>			
<u>Sesbania sp.</u>			
<u>Pithecolobium dulce</u>		Better dust collector.	Rao (1971)
<u>Argyrea speciosa</u>			
<u>Leucaena leucocephala</u>			
<u>Melilotus alba</u>	Polluted area		Ghouse and Khan (1983)
Banana Crop.	SO ₂ and dust From brick Kiln		Bedi <i>et al.</i> (1982)
<u>Lycopersicum esculentum</u>	SO ₂ and dust from brick Kiln		Beil and Bedi (1981)
<u>Mangifera indica</u>	SO ₂		Rao 1972 Shetye 1979 Giridhar (an published data) Pawar and Dubey (1983) Chaphekar <i>et al.</i> (1980 a)
<u>Helianthus annuus</u>	To pollute areas		
<u>Crotalaria juncea</u>			
<u>Commelina benghalensis</u>			
<u>Cyamopsis tetragonoloba</u>			
<u>Cicer arietinum</u>	Fly ash SO ₂		Dubey <i>et al.</i> (1982)

(Contd.....)

Table 2.6 (Contd.....)

Name of Plant	Sensitive	Tolerant	Reference
<u>Medicago sativa</u> (Alfa-alfa)	SO ₂		Singh and Rao (1979, 1980)
<u>Sorghum vulgare</u> var CSH - 1	SO ₂		Boralkar and Chaphekar (1978)
<u>Glycine max</u>	SO ₂		Pandey and Rao (1979), Prasad and Rao (1982)
<u>Phaseolus aureus</u>	SO ₂		Singh and Rao (1980)
<u>Arachis hypogea</u>	SO ₂		Mishra (1980)
<u>Dolichos lablab</u>	SO ₂		Banerjee and Chaphekar (1978)
<u>Phaseolus aureus</u> Var. Vaishakhee	SO ₂		Boralkar and Chaphekar (1980)
<u>Trigonella foenum-</u> <u>graecum</u>	SO ₂		Boralkar and Chaphekar (1983)
<u>Pisum sativum</u>	SO ₂		Varshney and Varshney (1978)
<u>Crossandra undulaefolia</u>	SO ₂		Chaphekar and Karbhari (1974)
<u>Mirabilis jalapa</u>			
<u>Amaranthus spinosus</u>	SO ₂		Boralkar and Chaphekar (1980)
<u>Spinacea oleracea</u>			
<u>Raphanus sativus</u>	SO ₂		Banerjee and Chaphekar (1978)
<u>Commelina benghalensis</u>			
<u>Erythrina indica</u>			
Barley, Cotton, Wheat, Aster, Cosmos, Verbena, Zinnia, Sweet Pea, Ipomoea purpurea, 4 o'clock plant, Bean Beet, Carrot, Chilli, Pumpkin, Radish Bhendi, Sunflower etc. Most trees	SO ₂		Pandey and Vaidya (1979)
		SO ₂	Pandey and Vaidya (1979)
<u>Mangifera indica</u>	SO ₂		Chaphekar (1972)
<u>Terminalia catappa</u>			
<u>Malachra capitata</u> Dahlia			
<u>Croton</u> , Plumeria		SO ₂	Chaphekar (1972)
Opuntia, Nerium			
Dahlia, Petunia, Alfalfa, cotton Barley	SO ₂		Vaishnavi (1975)

(Contd.....)

Table 2.6 (Contd.....)

Name of Plant	Sensitive	Tolerant	Reference
<u>Dalbergia sissoo</u>	SO ₂		Yunus and Ahmed (1979)
<u>Terminalia arjuna</u>			
<u>Cassia fistula</u>			
<u>Cedrela toona</u>			
<u>Syzygium cumini</u> Oat, Pea, Brinjal, Potato Cucurbit			
<u>Azadirachta indica</u>		SO ₂	Yunus and Ahmed (1979)
<u>Ficus religiosa</u>			
<u>Pithecolobium dulce</u>			
<u>Calotropis procera</u>			
Trees, Bushes, crops of those areas.			
<u>Phaseolus aureus</u>	SO ₂ , O ₃ , SO ₂ +O ₃		Agrawal and Rao (1983)
<u>Cicer arietinum</u>		SO ₂ , O ₃ , SO ₂ +O ₃	
<u>Oryza sativa</u>	SO ₂ , O ₃ , SO ₂ +O ₃		
<u>Panicum miliaceum</u>		SO ₃ , O ₃ , SO ₂ +O ₃	
<u>Solanum melongena</u>	SO ₂ , O ₃ , SO ₂ +O ₃		
<u>Vicia faba</u>	SO ₂ , O ₃ , SO ₂ +O ₃		
<u>Abelmoschus esculentus</u> , Var. Pusa savani	SO ₂ , O ₃ , SO ₂ +O ₃		
<u>Abelmoschus esculentus</u>	SO ₂ , O ₃ , SO ₂ +O ₃		Bolalkar and Shinde (1983)
<u>Phaseolus aureus</u>	SO ₂ , HF,		Sharma (1981)
<u>Triticum aestivum</u>	SO ₂ +HF		
<u>Brassica juncea</u>			
<u>Triticum aestivum</u>	NO ₂		Prasad and Rao (1979)
<u>Triticum aestivum</u>	NO ₂ +SO ₂		Prasad (1980)
<u>Dalbergia sissoo</u>	SO ₂		Rao <i>et al.</i> (1983)
<u>Madhuca indica</u>			
<u>Pisum sativum</u> var. Bonneville,	NaF		
<u>Pisum sativum</u> var. T163			
<u>Hordeum vulgare</u>			
<u>Zea mays</u>			
<u>Lycoopersicum esculentum</u>	NaF		Arya (1971)
<u>Terminalia tomentosa</u>	HF		Pandey (1979)
<u>Buchanania lanzan</u>			
<u>Zea mays</u>	HF		Rao and Pal (1978 b)
<u>Gladiolus</u> sp.	HF		Pandey and Rao (1980 a)

(Contd.....)

Table 2.6 (Contd....)

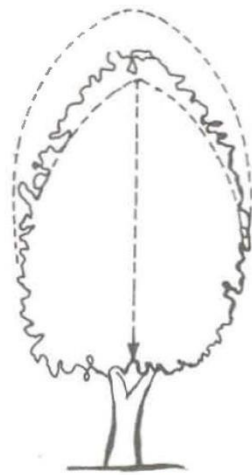
Name of Plant	Sensitive	Tolerant	Reference
<u>Spinacia oleracea</u>	Gasoline		Prasad (1980)
<u>Abelmoschus esculantus</u>	Vapour.		
<u>Cyamopsis tetragonoloba</u>	Ammonia		Chaphekar and Boralkar (1979)
<u>Crotalaria juncea</u>			
<u>Trigonella foenum-graecum</u>			
<u>Nerium indicum</u>	SO ₂		Varshney, (Unpublished)
<u>Cynodon dactylon</u>	HF		Meenakshy <u>et al</u> (1981)
<u>Cicer arietinum</u>	SO ₂		Varshney and Varshney (1981)
<u>Nasturtium indicum</u>			
<u>Petunia alba</u>			
<u>Tradescantia axillaris</u>			
<u>Madhuca indica</u>	SO ₂ , fly-ash		Agrawal M (1989)
<u>Cassia siamea</u>	-- " --		
<u>Delonix regia</u>			
<u>Shorea robusta</u>			
<u>Acacia arabica</u>		SO ₂ , fly-ash	
<u>Acacia catechu</u>			
<u>Zizyphus sp.</u>			
<u>Mangifera indica</u>		Dust	Agrawal & Khanam (1989)
<u>Ficus benghalensis</u> L		Dust	Ahmad Yunus <u>et al</u> (1991)
<u>Ficus infectoria</u> Roxb			
<u>Holoptelia integrifolia</u> Planch			
<u>Ipomoea fistulosa</u> Mart ex choisy			
<u>Lagerstroemia</u> sp.			
<u>Nyctanthes arbortristis</u> L			
<u>Peltophorum pterocarpum</u> (DC) K.Heyne			
<u>Tectona grandis</u> L		Dust	Ahmad Yunus <u>et al</u> (1991)
<u>Terminalia arjuna</u> W & A			
<u>Thevetia nerifolia</u> Jass			
<u>Acacia arabica</u> Willd			
<u>Bougainvillea spectabilis</u> Willd			
<u>Hibiscus rosa sinensis</u> Willd			
<u>Morus alba</u>			

(Contd....)

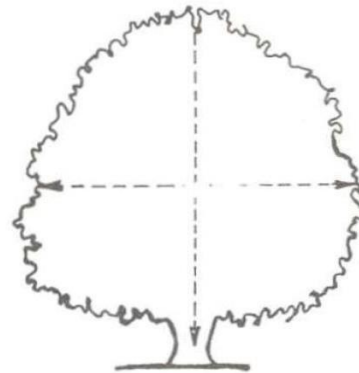
Table 2.6 (Contd....)

Name of Plant	Sensitive	Tolerant	Reference
<u>Nerium indicum</u> Mill		Cement dust	
<u>Thevetia nerifolia</u> juss			
<u>Dalbergia sissoo</u> Roxb			
<u>Azadirachta indica</u> A.Juss	Cement dust		Pandey, Misra et al (1994)
<u>Brassica campestris</u> L			
<u>Citrus aurantium</u> L			
<u>Delonix regia</u> Raffin			
<u>Syzygium cumini</u> (L) skeel			
<u>Mangifera indica</u> L			
<u>Pisum sativum</u> L			
<u>Tabernaemontana coronaria</u> Willd			
<u>Triticum aestivum</u> L			
<u>Zizyphus mauritiana</u> Lamk			
<u>Helianthus annuus</u> L		fly ash	
<u>Opuntia monacantha</u>		SO ₂	Raza S.H., Shylaja G. (1992)
<u>Opuntia dillenii</u>			
<u>Kalanchoe marmorata</u>			
<u>Crassula</u>			
<u>Bryophyllum</u>	SO ₂		
<u>Aloe</u>			
<u>Bryophyllum tabiflorum</u>			
<u>Euphorbia cattamindios</u>			
<u>Caesalpinia pulcherima</u>		SO ₂	Murthy M.S.R. et al (1990)
<u>Eugenia jambolana</u>			
<u>Polyalthia longifolia</u>	SO ₂		Raza S.H. et al (1991)
<u>Pongamia pinnata</u>			
<u>Caesalpinia pulcherima</u>		Dust	
<u>Pithecolobium dulce</u>			
<u>Cassia fistula</u>			
<u>Pongamia globra</u>	Dust		
<u>Polyalthia longifolia</u>			
<u>Pithecolobium dulce</u>		SO ₂	Raza S.H. et al (1989)
<u>Caesalpinia pulcherrima</u>			
<u>Polyalthia longifolia</u>	SO ₂		
<u>Pongamia pinnata</u>			

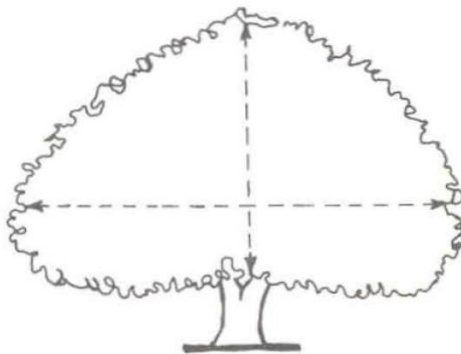
Fig.5.1 TREE CANOPY SHAPES



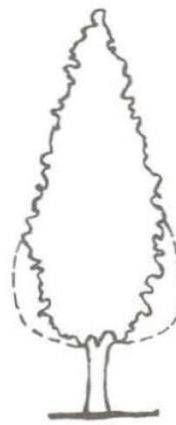
OBLONG



ROUND



SPREADING



CONICAL

The shapes given here are for convenience only. Many crown shapes range between those identified following, viz. Oblong-Round, Round-Spreading, Conical-Oblong, etc. Some shapes also change with age or environmental stresses.

FIG. 5.1 TREE CANOPY SHAPES

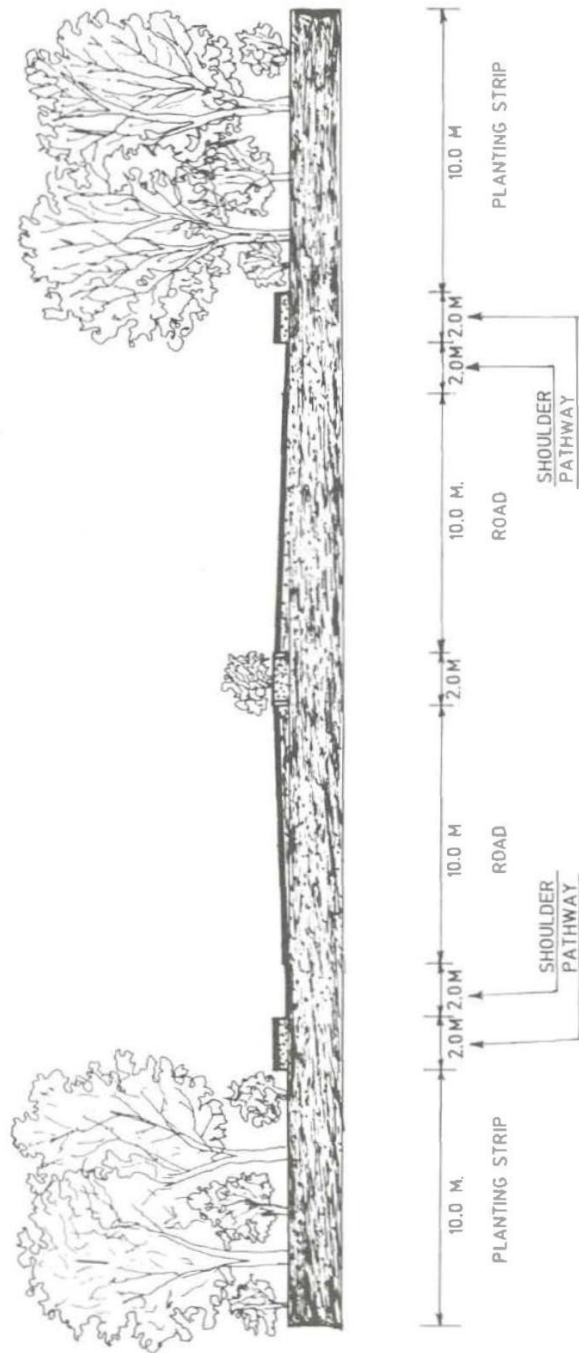


FIG. 5.2 TYPICAL ROAD-SIDE PLANTATION